Class 30

Suppose you have an strategic form game

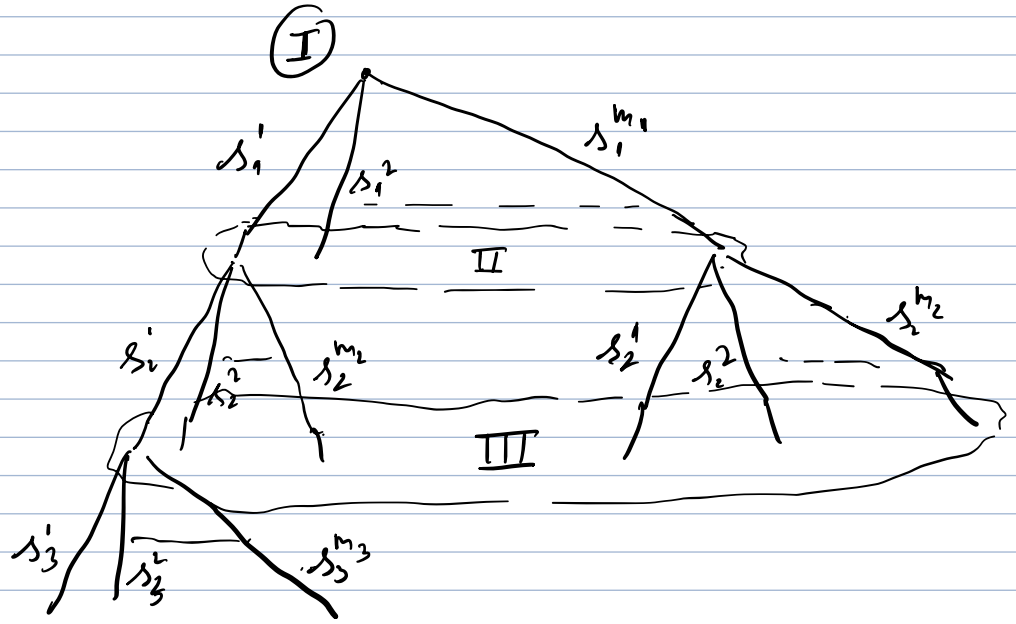
Question 1: Can you get an equivalent extensive form game?

$$G_1 \longrightarrow \Gamma \longrightarrow G_1 \quad (\text{equivalence})$$

Question 2: If exists, whether such an extensive form is unique or not?

Answer to question 1: $G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$

assume that $S_i = \{s_i^1, \dots, s_i^{m_i}\}$ $\forall i$



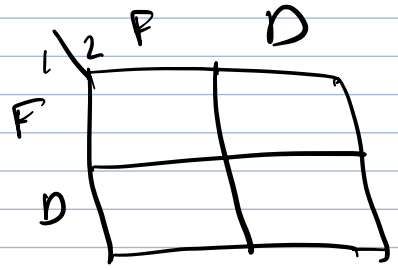
Every leaf node corresponds to a unique strategy profile in the strategic form game.

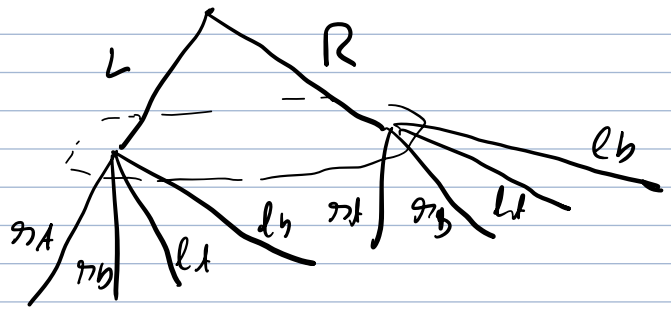
inside the utilities as those utilities.

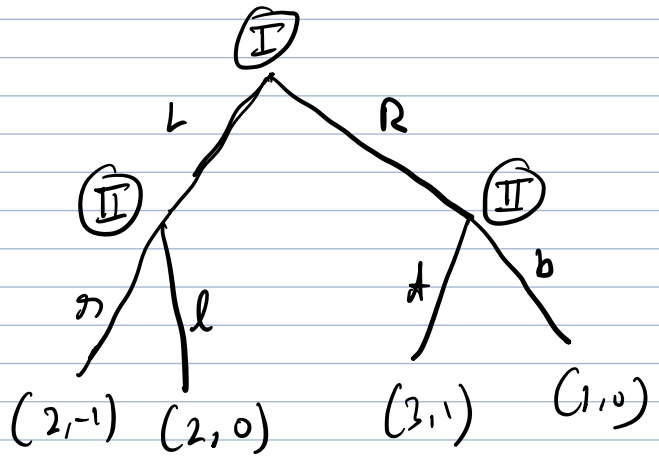
It is easy to see that the corresponding strategic form game of the above extensive form game is as follows:

If you change the ordering of the players in the extensive form game, you will get another equivalent form.

1 \ 2	$r_1 t$	$r_1 b$	$l t$	$l b$
L	(2,-1)	(2,-1)	(2,0)	(2,0)
R	(3,1)	(1,0)	(2,1)	(1,0)







Incomplete information games

This model is due to Harsanyi (1967)

Five 13 cards, we can calculate the probabilities of the cards other ones having.

$$\binom{52}{13}\binom{39}{13}\binom{26}{13}\checkmark$$

$$\binom{39}{13}\binom{26}{13}$$

An incomplete information game is a vector.

$$\langle N, (T_i)_{i\in N}, p, S, (S_A)_{A\in \underset{i\in N}{\times} T_i} \rangle$$

N: a finite set of players

$|T_i| = \binom{52}{13}$

T_i: a finite set of types for player i. The set of type vectors is denoted by $T = \underset{i\in N}{\times} T_i$

$|T| = \left[\binom{52}{13}\right]^4$

p: is a prob distribution over T. We assume that

$$p(t_i) = \sum_{t_{-i}\in T_{-i}} p(t_i, t_{-i}) > 0 \quad \forall i \in N \text{ all } t_i \in T_i$$

S: S is set of state games. Every state game $s \in S$ is a vector $s = (N, (A_i)_{i\in N}, (u_i)_{i\in N})$

where A_i is a non-empty set of actions available to player i. and $u_i : \times A_i \to \mathbb{R}$ is the utility function of the player i $i \in N$

s_A : is an element in S where

$$s_A = \left(N, \left(A_i(t_i)\right)_{i \in N}, \left(u_i(t)\right)_{i \in N} \right)$$

This is important to note that player i's action depends on his own type only.